

**Zoltán Szabó**  
Professor of Data Science

(Last updated: 11<sup>th</sup> September, 2026)

## CONTACT INFORMATION

Department of Statistics  
London School of Economics (LSE)  
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London, WC2A 2AE  
United Kingdom

Web: <https://zoltansz.github.io/>  
E-mail: x@y [x = z.szabo, y = lse.ac.uk]

## RESEARCH INTEREST

- Theory: statistical machine learning; kernel methods, information theory<sup>1</sup>, randomized algorithms, scalable computation.
- Applications:
  - hypothesis testing, shape-constrained prediction, safety-critical learning, style transfer, distribution regression, dictionary learning, structured sparsity, independent subspace analysis, bioinformatics, Bayesian inference, computer vision,
  - finance, economics, analysis of climate data, criminal data analysis, collaborative filtering, emotion recognition, face tracking, remote sensing, natural language processing, gene analysis.

## EMPLOYMENT

|                                                    |           |
|----------------------------------------------------|-----------|
| <b>London School of Economics</b> , United Kingdom |           |
| <b>Department of Statistics</b>                    |           |
| Professor of Data Science                          | 2021–     |
| <b>École Polytechnique</b> , France                |           |
| <b>Center of Applied Mathematics</b>               |           |
| Senior Researcher (1st class, HDR)                 | 2019–2021 |
| Senior Researcher (1st class)                      | 2016–2019 |
| <b>University College London</b> , United Kingdom  |           |
| <b>Gatsby Unit</b>                                 |           |
| Research Associate (with Prof. Arthur Gretton)     | 2013–2016 |
| <b>Eötvös Loránd University</b> , Hungary          |           |
| <b>School of Computer Science</b>                  |           |
| Research Fellow                                    | 2009–2013 |
| Assistant Research Fellow                          | 2008–2009 |
| Assistant Professor                                | 2007–2008 |

## EDUCATION AND HABILITATION

|                                                               |           |
|---------------------------------------------------------------|-----------|
| <b>Paris-Sud University</b> , France                          |           |
| HDR (with distinction)                                        | 2019      |
| Title: Contributions to Kernel Techniques.                    |           |
| <b>Eötvös Loránd University</b> , Hungary                     |           |
| <b>School of Computer Science</b>                             |           |
| Ph.D. (summa cum laude)                                       | 2004–2007 |
| Title: Separation Principles in Independent Process Analysis. |           |
| <b>Faculty of Natural Sciences, Applied Mathematics</b>       |           |

<sup>1</sup>Information Theoretical Estimators (ITE) Toolbox: <https://bitbucket.org/szzoli/ite-in-python/>.

|                                                                                                        |           |
|--------------------------------------------------------------------------------------------------------|-----------|
| Ph.D. (summa cum laude)                                                                                | 2003–2006 |
| Title: Group-Structured and Independent Subspace Based Dictionary Learning.                            |           |
| M.Sc. (with distinction)                                                                               | 1998–2003 |
| Title: Retina Based Sampling in Face Component Recognition.                                            |           |
| Specialization in Systems Theory, Signal and Image Processing,<br>Financial and Actuarial Mathematics. |           |

## PROFESSIONAL ACTIVITIES (GLOBAL)

### Moderator of

|                                                 |       |
|-------------------------------------------------|-------|
| Statistical Machine Learning (stat.ML) on arXiv | 2018– |
|-------------------------------------------------|-------|

### Reviewing Grant Applications

|                                          |       |
|------------------------------------------|-------|
| German Research Foundation (DFG)         | 2026– |
| Israel Science Foundation (ISF)          | 2021– |
| European Research Council (ERC)          | 2018– |
| Swiss National Science Foundation (SNSF) | 2017– |

### Senior Area Chair (SAC)

|                                                                                   |      |
|-----------------------------------------------------------------------------------|------|
| International Conference on Artificial Intelligence and Statistics (AISTATS-2027) | 2027 |
| Advances in Neural Information Processing Systems (NeurIPS)                       | 2026 |
| Conference on Uncertainty in Artificial Intelligence (UAI)                        | 2026 |
| International Conference on Machine Learning (ICML)                               | 2026 |
| International Conference on Artificial Intelligence and Statistics (AISTATS-2026) | 2025 |
| Advances in Neural Information Processing Systems (NeurIPS)                       | 2025 |
| International Conference on Artificial Intelligence and Statistics (AISTATS-2025) | 2024 |
| Advances in Neural Information Processing Systems (NeurIPS)                       | 2024 |
| International Conference on Artificial Intelligence and Statistics (AISTATS-2024) | 2023 |
| International Conference on Artificial Intelligence and Statistics (AISTATS-2023) | 2022 |
| Advances in Neural Information Processing Systems (NeurIPS)                       | 2019 |

### Area Chair (AC)

|                                                                                   |      |
|-----------------------------------------------------------------------------------|------|
| Conference on Uncertainty in Artificial Intelligence (UAI)                        | 2025 |
| International Conference on Machine Learning (ICML)                               | 2025 |
| Conference on Uncertainty in Artificial Intelligence (UAI)                        | 2024 |
| International Conference on Machine Learning (ICML)                               | 2024 |
| Conference on Learning Theory (COLT)                                              | 2024 |
| Advances in Neural Information Processing Systems (NeurIPS)                       | 2023 |
| International Conference on Machine Learning (ICML)                               | 2023 |
| Conference on Uncertainty in Artificial Intelligence (UAI)                        | 2023 |
| Conference on Learning Theory (COLT)                                              | 2023 |
| International Conference on Learning Representations (ICLR-2023)                  | 2022 |
| Advances in Neural Information Processing Systems (NeurIPS)                       | 2022 |
| International Conference on Machine Learning (ICML)                               | 2022 |
| Conference on Uncertainty in Artificial Intelligence (UAI)                        | 2022 |
| Conference on Learning Theory (COLT)                                              | 2022 |
| International Conference on Artificial Intelligence and Statistics (AISTATS-2022) | 2021 |
| International Conference on Learning Representations (ICLR-2022)                  | 2021 |
| Advances in Neural Information Processing Systems (NeurIPS)                       | 2021 |
| International Conference on Machine Learning (ICML)                               | 2021 |
| Conference on Learning Theory (COLT)                                              | 2021 |
| Conference on Uncertainty in Artificial Intelligence (UAI)                        | 2021 |
| International Joint Conference on Artificial Intelligence (IJCAI)                 | 2021 |
| International Conference on Artificial Intelligence and Statistics (AISTATS-2021) | 2020 |

|                                                                                   |           |
|-----------------------------------------------------------------------------------|-----------|
| International Conference on Learning Representations (ICLR-2021)                  | 2020      |
| Advances in Neural Information Processing Systems (NeurIPS)                       | 2020      |
| Conference on Uncertainty in Artificial Intelligence (UAI)                        | 2020      |
| International Conference on Machine Learning (ICML)                               | 2020      |
| International Joint Conference on Artificial Intelligence (IJCAI)                 | 2020      |
| International Conference on Artificial Intelligence and Statistics (AISTATS-2020) | 2019      |
| International Joint Conference on Artificial Intelligence (IJCAI)                 | 2019      |
| International Conference on Machine Learning (ICML)                               | 2019      |
| International Conference on Artificial Intelligence and Statistics (AISTATS-2019) | 2018      |
| Advances in Neural Information Processing Systems (NeurIPS)                       | 2018      |
| International Conference on Machine Learning (ICML)                               | 2018      |
| International Conference on Artificial Intelligence and Statistics (AISTATS-2018) | 2017      |
| Conference on Uncertainty in Artificial Intelligence (UAI)                        | 2017      |
| International Conference on Machine Learning (ICML)                               | 2017      |
| International Conference on Artificial Intelligence and Statistics (AISTATS-2017) | 2016      |
| Conference on Uncertainty in Artificial Intelligence (UAI)                        | 2016      |
| <b>Advisory Committee</b>                                                         |           |
| arXiv Statistics Advisory Committee Member                                        | 2019–     |
| <b>Editorial Board Member</b>                                                     |           |
| Journal of Machine Learning Research                                              | 2020–     |
| <b>Senior Associate Editor</b>                                                    |           |
| ACM Transactions on Probabilistic Machine Learning                                | 2023–     |
| <b>Associate Editor</b>                                                           |           |
| Mathematical Foundations of Computing                                             | 2019–     |
| <b>Reviewing for Journals</b>                                                     |           |
| SIAM/ASA Journal on Uncertainty Quantification                                    | 2026–     |
| Journal of the Royal Statistical Society: Series B                                | 2024–     |
| Annals of the Institute of Statistical Mathematics                                | 2024–     |
| Constructive Approximation                                                        | 2023–     |
| Latin American Journal of Probability and Mathematical Statistics                 | 2022–     |
| Applied and Computational Harmonic Analysis                                       | 2022–     |
| Journal of Complexity                                                             | 2022–     |
| Annals of Applied Probability                                                     | 2021–     |
| Journal of the American Statistical Association                                   | 2021–     |
| Information and Inference: A Journal of the IMA                                   | 2020–     |
| Nature Machine Intelligence                                                       | 2020–     |
| Transactions on Knowledge and Data Engineering                                    | 2020–     |
| Foundations of Data Science                                                       | 2019–     |
| Foundations of Computational Mathematics                                          | 2019–     |
| Electronic Journal of Statistics                                                  | 2019–     |
| Journal of Multivariate Analysis                                                  | 2017–     |
| Annals of Statistics                                                              | 2016–     |
| Machine Learning                                                                  | 2016–     |
| Statistics and Computing                                                          | 2015–     |
| IEEE Signal Processing Letters                                                    | 2015–     |
| Statistical Analysis and Data Mining                                              | 2014–     |
| IET Computer Vision                                                               | 2014–     |
| International Journal of Computer Vision                                          | 2014–     |
| IEEE Transactions on Information Theory                                           | 2013–     |
| Journal of Machine Learning Research                                              | 2013–2019 |

|                                                                                                                             |                       |
|-----------------------------------------------------------------------------------------------------------------------------|-----------------------|
| IEEE Transactions on Pattern Analysis and Machine Intelligence                                                              | 2013–                 |
| Progress in Artificial Intelligence                                                                                         | 2013–                 |
| Entropy                                                                                                                     | 2012–                 |
| IEEE Transactions on Neural Networks and Learning Systems                                                                   | 2012–                 |
| Signal, Image and Video Processing                                                                                          | 2012–                 |
| IEEE Transactions on Signal Processing                                                                                      | 2009–                 |
| Neurocomputing                                                                                                              | 2009–                 |
| IEEE Transactions on Neural Networks                                                                                        | 2007–2011             |
| <b>Reviewing Books</b>                                                                                                      |                       |
| Cambridge University Press                                                                                                  | 2018–                 |
| <b>Reviewing for Conferences</b>                                                                                            |                       |
| IEEE International Symposium on Information Theory (ISIT)                                                                   | 2021                  |
| International Conference on Artificial Neural Networks (ICANN)                                                              | 2019                  |
| International Conference on Learning Representations (ICLR-2019)                                                            | 2018                  |
| Conference on Learning Theory (COLT)                                                                                        | 2018                  |
| International Conference on Learning Representations (ICLR-2018)                                                            | 2017                  |
| Conference on Learning Theory (COLT)                                                                                        | 2017                  |
| Advances in Neural Information Processing Systems (NIPS)                                                                    | 2017                  |
| Advances in Neural Information Processing Systems (NIPS)                                                                    | 2016                  |
| International Conference on Machine Learning (ICML)                                                                         | 2016                  |
| Advances in Neural Information Processing Systems (NIPS)                                                                    | 2015                  |
| Advances in Neural Information Processing Systems (NIPS)                                                                    | 2014                  |
| International Conference on Machine Learning (ICML)                                                                         | 2012                  |
| International Conference on Latent Variable Analysis and Signal Separation (LVA/ICA)                                        | 2012                  |
| International Joint Conference on Artificial Intelligence (IJCAI)                                                           | 2011                  |
| International Joint Conference on Neural Networks (IJCNN)                                                                   | 2011                  |
| European Conference on Complex Systems (ECCS)                                                                               | 2011                  |
| European Signal Processing Conference (EUSIPCO)                                                                             | 2011                  |
| European Conference on Complex Systems (ECCS)                                                                               | 2009                  |
| <b>Reviewing Workshop Proposals</b>                                                                                         |                       |
| International Conference on Machine Learning (ICML) Workshops                                                               | 2019                  |
| <b>Reviewing for Workshops</b>                                                                                              |                       |
| Signal Processing with Adaptive Sparse Structured Representations (SPARS)                                                   | 2019                  |
| Challenges in Machine Learning (CiML) @ N(eur)IPS:                                                                          |                       |
| Machine Learning Competitions “in the Wild”: Playing in the Real World or in Real                                           | 2018                  |
| Time Gaming and Education                                                                                                   | 2016                  |
| <b>Organizing (workshop, conference)</b>                                                                                    |                       |
| Representing, calibrating & leveraging prediction uncertainty from statistics to machine learning (Isaac Newton Institute): | May 6 - Aug. 29, 2025 |
| W1 Uncertainty in multivariate, non-Euclidean, and functional spaces: theory and practice,                                  | May 6-9, 2025         |
| – principal workshop organizer,                                                                                             |                       |
| – co-organizers: Alexandra Menafoglio, David Ginsbourger, Florence d’Alché-Buc, Judith Rousseau, Neil Lawrence.             |                       |
| NIPS: ‘Learning on Distributions, Functions, Graphs and Groups’                                                             | 2017                  |
| – co-organizers: Bharath K. Sriperumbudur, Florence d’Alché-Buc, Krikamol Muandet.                                          |                       |
| NIPS: ‘Adaptive and Scalable Nonparametric Methods in Machine Learning’                                                     | 2016                  |
| – co-organizers: Aaditya Ramdas, Bharath K. Sriperumbudur, Han Liu, John Lafferty, Mladen Kolar, Samory Kpotufe.            |                       |

|                                                                                                                                                                                                                      |               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| NIPS: ‘Modern Nonparametrics 3: Automating the Learning Pipeline’<br>– co-organizers: Andrew G. Wilson, Arthur Gretton, Eric Xing, Han Liu,<br>Le Song, Mladen Kolar, Samory Kpotufe.                                | 2014          |
| Conference (Workflow Chair)<br>International Conference on Artificial Intelligence and Statistics (AISTATS)<br>– co-workflow chair: Rodolphe Jenatton.                                                               | 2016          |
| <b>Session Chair (conference)</b>                                                                                                                                                                                    |               |
| Lifting Inference with Kernel Embeddings (LIKE)<br>Session: Kernel-based Hypothesis Testing and Optimization                                                                                                         | 2023          |
| International Conference on Artificial Intelligence and Statistics (AISTATS)<br>Session: Learning Theory / Kernels.                                                                                                  | 2022          |
| International Conference on Machine Learning (ICML)<br>Sessions: Kernel Methods, Learning Theory.                                                                                                                    | 2021          |
| Conference on Uncertainty in Artificial Intelligence (UAI)<br>Session: Optimization.                                                                                                                                 | 2020          |
| International Conference on Mathematics of Data Science (MathoDS 3)                                                                                                                                                  | 2019          |
| International Conference on Machine Learning (ICML)<br>Session: Statistical Learning Theory.                                                                                                                         | 2018          |
| Paris Summit on Big Data (ParisBD)                                                                                                                                                                                   | 2017          |
| <b>Mentoring Newcomers</b>                                                                                                                                                                                           |               |
| 3 Newcomers @ ICML                                                                                                                                                                                                   | 2020          |
| <b>Mentoring Oral Presentations</b>                                                                                                                                                                                  |               |
| Rahul Singh (MIT Economics) @ NeurIPS                                                                                                                                                                                | 2019          |
| <b>Reviewing Scientific Competitions</b>                                                                                                                                                                             |               |
| National Scientific Student Competition and Conference                                                                                                                                                               | 2005, 2013    |
| Scientific Student Competition and Conference                                                                                                                                                                        | 2012          |
| <b>Thesis Committee (Ph.D.)</b>                                                                                                                                                                                      |               |
| Yanzhi Chen (Ph.D.)<br>– Department of Engineering, University of Cambridge, UK,<br>– thesis title: On Dependence Modeling with Mutual Information,<br>– role: Reviewer.                                             | Apr. 13, 2026 |
| Masha Naslidnyk (Ph.D.)<br>– Department of Statistical Science, University College London, UK,<br>– thesis title: Scalable Kernel-Based Distances for Statistical Inference<br>and Integration,<br>– role: Reviewer. | Feb. 10, 2026 |
| Tobias Schröder (Ph.D.)<br>– Department of Mathematics, Imperial College London, UK,<br>– thesis title: New Estimation Methods for Energy-Based Models and<br>some Applications,<br>– role: Reviewer.                | Feb. 6, 2026  |
| Florian Kalinke (Ph.D.)<br>– Faculty of Computer Science, Karlsruhe Institute of Technology, Germany,<br>– role: Reviewer.                                                                                           | May 19, 2025  |
| Théophile Cantelobre (Ph.D.)<br>– SIERRA Team, INRIA Paris, France,<br>– thesis title: Contributions to Structured Statistical Learning:<br>Theory and Algorithms,<br>– role: Examiner.                              | Oct. 16, 2024 |
| Tamim El Ahmad (Ph.D.)                                                                                                                                                                                               | July 9, 2024  |

- LTCI, Télécom Paris, France.
- thesis title: Learning Deep Kernel Networks: Application to Efficient and Robust Structured Prediction,
- role: Examiner.
- Omar Hagrass (Ph.D.) May 20, 2024
  - Department of Statistics, Pennsylvania State University, US,
  - thesis title: Spectral Regularized Kernel Two-Sample Tests,
  - role: Reviewer.
- Anthony Ozier-Lafontaine (Ph.D.) Nov. 24, 2023
  - Jean Leray Mathematics Lab, University of Nantes, France,
  - thesis title: Kernel-based testing and their application to single-cell data,
  - role: Reviewer.
- Zhanliang Huang (Ph.D.) May 30, 2023
  - School of Computer, University of Birmingham, UK,
  - thesis title: Noise Reduction in Differentially Private Learning,
  - role: Reviewer.
- Hai Pham (Ph.D.) Apr. 28, 2023
  - School of Computer Science, Carnegie Mellon University, US,
  - thesis title: Towards Efficient and Scalable Representation Learning,
  - role: Reviewer.
- Valerii Likhoshervostov (Ph.D.) Mar. 15, 2023
  - Department of Engineering, University of Cambridge, UK,
  - thesis title: Random Features for Efficient Attention Approximation,
  - role: Reviewer.
- Omar Hagrass (Ph.D.) Nov. 15, 2022
  - Department of Statistics, Pennsylvania State University, US,
  - thesis title: Spectral Regularized Kernel Two-Sample Tests,
  - role: Reviewer of Comprehensive Exam.
- Jonas Wacker (Ph.D.) July 12, 2022
  - Data Science Department, Eurecom, France,
  - thesis title: Random Features for Dot Product Kernels and Beyond,
  - role: Reviewer.
- Hai Pham (Ph.D.) Apr. 21, 2022
  - School of Computer Science, Carnegie Mellon University, US,
  - thesis title: Towards Efficient and Scalable Representation Learning,
  - role: Committee Member for Thesis Proposal.
- Guillaume Staerman (Ph.D.) Apr. 12, 2022
  - Télécom Paris, France,
  - thesis title: Functional Anomaly Detection and Robust Estimation,
  - role: Reviewer.
- Luigi Carratino (Ph.D.) 2020
  - Computer Science and Systems Engineering Program, University of Genova, Italy,
  - thesis title: Resource Efficient Large-Scale Machine Learning,
  - role: Reviewer.
- Romain Brault (Ph.D.) 2017
  - Télécom ParisTech, France,
  - thesis title: Large-scale Operator-Valued Kernel Regression,
  - role: Reviewer.
- Kornél Kovács (Ph.D.) 2008
  - University of Szeged, Hungary,
  - thesis title: Various Kernel Methods with Applications,
  - role: Reviewer.

**Thesis Committee (M.Sc.)**

|                                                                                                  |               |
|--------------------------------------------------------------------------------------------------|---------------|
| Mialimanana Andrianina Bien-aimé Razafindrakoto (M.Sc.)                                          | June 24, 2026 |
| – AI for Science program, African Institute for Mathematical Sciences (AIMS), Cape Town, Africa. |               |
| – thesis title: Quasi-Monte Carlo Sampling for Kernel Quantile Embeddings,                       |               |
| – role: Reviewer.                                                                                |               |
| Gábor Matuz (M.Sc.)                                                                              | 2010          |
| – Budapest University of Technology and Economics, Hungary,                                      |               |
| – thesis title: Adaptive Algorithms in Multiagent Environments,                                  |               |
| – role: Reviewer.                                                                                |               |

**PROFESSIONAL ACTIVITIES (LONDON SCHOOL OF ECONOMICS)****Programme Director**

|                  |              |
|------------------|--------------|
| MSc Data Science | Sept., 2022– |
|------------------|--------------|

**Sub-Board Chair of MSc Data Science**

Sept., 2022–

**Management Committee/Board**

|                          |             |
|--------------------------|-------------|
| DTS Management Board     | June, 2024– |
| DSI Management Committee | 2021–2025   |

**Turing Academic Liaison**

2023–2025

**Member of Review Panel**

|                            |                 |
|----------------------------|-----------------|
| RISF AI grant applications | Feb.-Mar., 2024 |
|----------------------------|-----------------|

**Recruitment Committee**

|                               |            |
|-------------------------------|------------|
| Departmental Search Committee | 2022–2025  |
| Departmental Research Panel   | 2022–2025  |
| Departmental Teaching Panel   | 2022–2024  |
| Departmental Selection Panel  | Mar., 2024 |
| LSE Fellow Recruitment Panel  | Apr., 2022 |

**Departmental EDI Committee**

Sept., 2023–

**Departmental Research Committee**

Sept., 2021–

**Departmental Teaching Committee**

Sept., 2021–

**Organizing (research showcase)**

|                                                                                |                  |
|--------------------------------------------------------------------------------|------------------|
| Statistics Research Showcase                                                   | May 26-28, 2026  |
| co-organizers: Milan Vojnovic, Marcos Barreto,                                 |                  |
| PSS support: Penny Montague, Penelope Smith, John Barlow,                      |                  |
| Daniel Titherington.                                                           |                  |
| Statistics Research Showcase                                                   | Apr. 7–8, 2025   |
| PSS support: Penny Montague.                                                   |                  |
| Statistics Research Showcase                                                   | June 20–21, 2024 |
| co-organizers: Giulia Livieri, Sara Geneletti, Chengchun Shi, Kostas Kardaras, |                  |
| PSS support: Penny Montague.                                                   |                  |
| Statistics Research Showcase                                                   | June 5–6, 2023   |
| co-organizers: Yunxiao Chen, Tengyao Wang, Erik Baurdoux,                      |                  |
| PSS support: Penny Montague, Charlotte Morgan.                                 |                  |
| Statistics Research Showcase                                                   | June 14–15, 2022 |
| co-organizers: Fiona Steele, Clifford Lam, Kostas Kardaras,                    |                  |
| PSS support: Penny Montague, Joey Hoang.                                       |                  |

**Organizing (seminar)**

|                                                                                                                                         |                        |
|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Industry Speakers Seminar @ Department of Statistics                                                                                    | Apr. 2025–Aug. 2026    |
| Data Science Seminar @ Department of Statistics                                                                                         | 2021–2023              |
| co-organizer: Francesca Panero.                                                                                                         | 2022–2023              |
| <b>Organizing (Cumberland Lodge weekend for MSc students)</b>                                                                           | Oct. 11-13, 2024       |
| co-organizer: Irini Moustaki.                                                                                                           |                        |
| <b>PhD Upgrade (MPhil to PhD) Reviewer</b>                                                                                              |                        |
| Xianghe Zhu (MPhil)                                                                                                                     | July 16, 2025          |
| – Department of Statistics, LSE,                                                                                                        |                        |
| – title: Autoregressive Hypergraphs,                                                                                                    |                        |
| – supervisor: Prof. Qiwei Yao.                                                                                                          |                        |
| Sakina Hansen (MPhil)                                                                                                                   | Sept. 26, 2024         |
| – Department of Statistics, LSE,                                                                                                        |                        |
| – title: Explainable Machine Learning: Philosophy, Methods and Applications,                                                            |                        |
| – supervisor: Prof. Joshua Loftus.                                                                                                      |                        |
| <b>System Administration</b>                                                                                                            | May–, 2024             |
| Departmental GPU server                                                                                                                 |                        |
| <b><math>\beta</math>-Testing and Shaping LSE RONIN</b>                                                                                 | July-Sept. 2025        |
| Working with the Research Computing team                                                                                                |                        |
| <b>2nd Examiner (ST = Spring Term, AT = Autumn Term)</b>                                                                                |                        |
| Capstone Project (ST498)                                                                                                                | ST, 2026               |
| Artificial Intelligence (ST311)                                                                                                         | ST, 2026               |
| Artificial Intelligence (ST449)                                                                                                         | AT, 2025               |
| Capstone Project (ST498)                                                                                                                | ST, 2025               |
| Artificial Intelligence (ST311)                                                                                                         | ST, 2025               |
| Artificial Intelligence (ST449)                                                                                                         | AT, 2024               |
| Capstone Project (ST498)                                                                                                                | ST, 2024               |
| Artificial Intelligence (ST311)                                                                                                         | ST, 2024               |
| Artificial Intelligence (ST449)                                                                                                         | AT, 2023               |
| Capstone Project (ST498)                                                                                                                | ST, 2023               |
| Artificial Intelligence (ST311)                                                                                                         | ST, 2023               |
| Capstone Project (ST498)                                                                                                                | ST, 2022               |
| Artificial Intelligence (ST311)                                                                                                         | ST, 2022               |
| <b>PROFESSIONAL ACTIVITIES (ÉCOLE POLYTECHNIQUE)</b>                                                                                    |                        |
| <b>Program Chair @ Data Science Summer School</b>                                                                                       |                        |
| co-organizers: Aldjia Mazari, Bertrand Thirion, Emmanuel Gobet, Erwan Le Pennec, Jasmyn Scaramella.                                     | 2020                   |
| co-organizers: Aldjia Mazari, Anastasiia Nitavskiyi, Aurélie Coen, Bertrand Thirion, Charlotte Renauld, Elena Carvajal, Emmanuel Bacry. | 2019                   |
| co-organizers: Aldjia Mazari, Charlotte Renauld, Emmanuel Bacry, Erwan Scornet, Maud Cadiz-Pena, Nozha Boujemaa, Viviane Hoang.         | 2018                   |
| co-organizers: Aldjia Mazari, Emmanuel Bacry, Éric Moulines, Erwan Scornet.                                                             | 2017                   |
| <b>Organizing (seminar)</b>                                                                                                             |                        |
| Statistics Seminars (CREST-CMAP)                                                                                                        | 2020–2021              |
| co-organizers: Cristina Butucea, Alexandre Tsybakov, Karim Lounici.                                                                     |                        |
| Machine Learning External Seminars @ CMAP                                                                                               | 2017–2021              |
| <b>Organizing (workshop)</b>                                                                                                            |                        |
| StressTest-2020 workshop                                                                                                                | Nov. 30 – Dec. 1, 2020 |

- co-organizers: Stefano De Marco, Emmanuel Gobet.
- topics: financial stress-testing, uncertainty quantification, risk and dependence modelling.

### Organizing (journal club, group meeting)

|                                                                       |           |
|-----------------------------------------------------------------------|-----------|
| SIMPAS Group Meeting @ CMAP                                           | 2020–2021 |
| – co-organizers: Aymeric Dieuleveut, Emmanuel Gobet, Erwan Le Pennec. |           |
| Machine Learning Journal Club @ CMAP                                  | 2017–2021 |

### Recruitment Committee

|                                          |               |
|------------------------------------------|---------------|
| Data Science for Business (M.Sc., X-HEC) | June 28, 2021 |
| Data Science for Business (M.Sc., X-HEC) | Apr. 1, 2020  |
| Data Science for Business (M.Sc., X-HEC) | May 27, 2019  |
| Data Science for Business (M.Sc., X-HEC) | Feb. 8, 2019  |
| Data Science for Business (M.Sc., X-HEC) | Nov. 21, 2018 |

### Internship Committee

|                                                                 |                 |
|-----------------------------------------------------------------|-----------------|
| X-HEC (M.Sc.)                                                   | Sept. 10, 2018  |
| Statistical Models in Biology and Physics (M.Sc.)               | Mar. 22, 2018   |
| Data Science (M.Sc.)                                            | Sept. 27, 2017  |
| – École Polytechnique (morning) & Télécom ParisTech (afternoon) |                 |
| Data Science (M.Sc.)                                            | Sept. 4–5, 2017 |

## PROFESSIONAL ACTIVITIES (UNIVERSITY COLLEGE LONDON)

### Organizing (seminar)

|                                                  |           |
|--------------------------------------------------|-----------|
| Machine Learning External Seminars @ Gatsby Unit | 2014–2016 |
|--------------------------------------------------|-----------|

## PUBLICATIONS

### Preprints

- [1] Jose Cribeiro-Ramallo, Florian Kalinke, and Zoltán Szabó. Minimax lower bounds of kernel discrepancy estimation: MMD, HSIC, KSD. Technical report, 2026. (<https://arxiv.org/abs/2607.24235>).
- [2] Florian Kalinke, Zoltán Szabó, and Bharath K. Sriperumbudur. Nyström kernel Stein discrepancy tests. Technical report, 2026. (<http://arxiv.org/abs/2605.25173>).
- [3] Linda Chamakh and Zoltán Szabó. Keep it tighter – a story on analytical mean embeddings. Technical report, 2024. (<https://arxiv.org/abs/2110.09516>).
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### Unreferred Conference, Workshop and Symposium Papers

- [1] Alex Lambert, Sanjeel Parekh, Zoltán Szabó, and Florence d’Alché-Buc. Continuous emotion transfer using kernels. In *Advances in Neural Information Processing Systems (NeurIPS): Workshop on Controllable Generative Modeling in Language and Vision (CtrlGen)*, 13 December 2021. (acceptance rate  $< 50\%$ ).
- [2] Pierre-Cyril Aubin-Frankowski and Zoltán Szabó. Hard shape-constrained kernel regression. In *SMAI-MODE*, 7-9 September 2020.
- [3] Pierre-Cyril Aubin-Frankowski and Zoltán Szabó. Hard shape-constrained kernel regression. In *Joint Structures and Common Foundations of Statistical Physics, Information Geometry and Inference for Learning (SPIGL)*, 27-31 July 2020.
- [4] Matthieu Lerasle, Zoltán Szabó, Timothée Mathieu, and Guillaume Lécué. Median-of-means for outlier-robust MMD estimation. In *Workshop on Data, Learning and Inference (DALI)*, San Sebastian, Spain, 2-5 September 2019.
- [5] Romain Brault<sup>†</sup>, Alex Lambert<sup>†</sup>, Zoltán Szabó, Maxime Sangnier, and Florence d’Alché-Buc. Infinite task learning in RKHSs. In *Conference on Machine Learning (CAp)*, Toulouse, France, 3-5 July 2019. (<sup>†</sup>contributed equally).
- [6] Alex Lambert<sup>†</sup>, Romain Brault<sup>†</sup>, Zoltán Szabó, Maxime Sangnier, and Florence d’Alché-Buc. A functional extension of multi-output learning. In *International Conference on Machine Learning (ICML): Adaptive & Multitask Learning (AMTL)*, Long Beach, CA, U.S., 15 June 2019. (<sup>†</sup>contributed equally).
- [7] Romain Brault<sup>†</sup>, Alex Lambert<sup>†</sup>, Zoltán Szabó, Maxime Sangnier, and Florence d’Alché-Buc. Infinite task learning. In *PASADENA Workshop*, Paris, France, 15 February 2019. (<sup>†</sup>contributed equally).
- [8] Zoltán Szabó and Bharath K. Sriperumbudur. Random Fourier features on kernel derivatives. In *Data Learning and Inference (DALI)*, George, South Africa, 3-5 January 2019.
- [9] Zoltán Szabó and Bharath K. Sriperumbudur. Independence via cross-covariance operators. In *Polish-Italian Mathematical Conference: Challenges and Methods of Modern Statistics*, Wroclaw, Poland, 17-20 September 2018.
- [10] Alex Lambert, Romain Brault, Zoltán Szabó, Maxime Sangnier, and Florence d’Alché-Buc. Infinite-task learning with vector-valued reproducing kernel Hilbert spaces. In *Junior Conference on Data Science and Engineering (JDSE)*, 13-14 September 2018.
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- [22] Wittawat Jitkrittum, Zoltán Szabó, Kacper Chwialkowski, and Arthur Gretton. Distinguishing distributions with interpretable features. In *International Conference on Machine Learning (ICML): Data-Efficient Machine Learning workshop*, New York, 24 June 2016.
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- [38] Zoltán Szabó. Independent subspace analysis in case of missing observations. In *Symposium of Intelligent Systems*, 20 November 2009.
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## Habilitation

- [1] Zoltán Szabó. Contributions to kernel techniques, 2019. (HDR, with distinction).

## Patent

- [1] Zoltán Szabó, László Jeni, and Dániel Takács. Method and apparatus with deformable model fitting using high-precision approximation. European Patent (EP2672425A1), 2013.

## Theses

- [1] Zoltán Szabó. *Group-Structured and Independent Subspace Based Dictionary Learning*. PhD thesis, Eötvös Loránd University, Budapest, 2012. (PhD in Applied Mathematics).
- [2] Zoltán Szabó. *Separation Principles in Independent Process Analysis*. PhD thesis, Eötvös Loránd University, Budapest, 2009. (PhD in Computer Science).
- [3] Zoltán Szabó. Retina based sampling in face component recognition. Master’s thesis, Eötvös Loránd University, Budapest, 2003.

## Technical Reports (Older)

- [1] Alex Lambert<sup>†</sup>, Sanjeel Parekh<sup>†</sup>, Zoltán Szabó, and Florence d’Alché-Buc. Emotion transfer using vector-valued infinite task learning. Technical report, 2021. (<sup>†</sup>contributed equally; <https://arxiv.org/abs/2102.05075>).
- [2] András Lőrincz, Viktor Gyenes, Zsolt Palotai, Balázs Pintér, Zoltán Szabó, and Gyula Vörös. Innovation engine for blogspaces (EOARD - US Air Force Research Laboratories). Technical report, Eötvös Loránd University, Budapest, 2011. (<http://www.dtic.mil/cgi-bin/GetTRDoc?AD=ADA550367>).
- [3] Zoltán Szabó. Towards nonstationary, nonparametric independent process analysis with unknown source component dimensions. Technical report, Eötvös Loránd University, Budapest, 2010. (<http://arxiv.org/abs/1008.1393>).
- [4] Zoltán Szabó, Barnabás Póczos, and András Lőrincz. Separation theorem for  $\mathbb{K}$ -independent subspace analysis with sufficient conditions. Technical report, Eötvös Loránd University, Budapest, 2006. (<http://arxiv.org/abs/math.ST/0608100>).
- [5] Zoltán Szabó, Barnabás Póczos, and András Lőrincz. Separation theorem for independent subspace analysis with sufficient conditions. Technical report, Eötvös Loránd University, Budapest, 2006. (<http://arxiv.org/abs/math.ST/0603535>).
- [6] Zoltán Szabó, Barnabás Póczos, and András Lőrincz. Separation theorem for independent subspace analysis. Technical report, Eötvös Loránd University, Budapest, 2005. ([https://zoltansz.github.io/publications/szabo05separation\\_TR.pdf](https://zoltansz.github.io/publications/szabo05separation_TR.pdf)).
- [7] Zoltán Szabó and András Lőrincz.  $L_1$  regularization is better than  $L_2$  for learning and predicting chaotic systems. Technical report, Eötvös Loránd University, Budapest, 2004. (<http://arxiv.org/abs/cs/0410015>).

## INVITED TALKS (GLOBAL)

- [1] Accelerated computation and minimax lower bound of kernel Stein discrepancy. Faculty of Engineering, Free University of Bozen-Bolzano, presentation (1 hour), 25 February 2026.
- [2] Mini-course on kernel techniques. Faculty of Engineering, Free University of Bozen-Bolzano, presentation (1 hour), 23 February 2026.
- [3] Minimax lower bound for kernel Stein discrepancy estimation. Numerical Mathematics and Data Science Seminar, School of Mathematics, University of Edinburgh, presentation (1 hour), 12 November 2025.

- [4] The minimax lower bound of kernel Stein discrepancy estimation. UCL JumpTrading/ELLIS CSML Seminar Series, Gatsby Unit, University College London, presentation (1 hour), 5 November 2025.
- [5] Kernel cumulants. Data Science Seminar, University of York, presentation (1 hour), 30 April 2025.
- [6] The power of cumulants in reproducing kernel Hilbert spaces. Computer Science Colloquium, Department of Computer Science, University of Warwick, UK, presentation (1 hour), 12 February 2024.
- [7] Nyström M-HSIC. CMStatistics, Economic Data Analysis and Statistical Inference to Unfold Uncertainty session, Berlin, Germany, presentation (25 minutes), 18 December 2023.
- [8] Beyond mean embedding: Cumulants in RKHSs. CMStatistics, Statistical Machine Learning with Kernels and Non-linear Transformations session, Berlin, Germany, presentation (25 minutes), 17 December 2023.
- [9] Kernel cumulants. CMStatistics, Advances in kernel methods and Gaussian processes session, Berlin, Germany, presentation (25 minutes), 16 December 2023.
- [10] Kernel cumulant embedding. Lifting Inference with Kernel Embeddings (LIKE23), Bern, Switzerland, presentation (45 minutes), 28 June 2023.
- [11] Beyond mean embedding: The power of cumulants in RKHSs. Advanced Course on Data Science and Machine Learning (ACDL), Tuscany, Italy, plenary talk (45 minutes), 11 June 2023.
- [12] Shape-constrained kernel machines and their applications. Advanced Course on Data Science and Machine Learning (ACDL), Tuscany, Italy, plenary talk (45 minutes), 11 June 2023.
- [13] Kernel machines with shape constraints. BIRS workshop on New Interfaces of Stochastic Analysis and Rough Paths, presentation (25 minutes), 8 September 2022.
- [14] When shape constraints meet kernel machines. International Conference on Econometrics and Statistics (EcoSta): Recent Advances in Machine Learning session, presentation (25 minutes), 4 June 2022.
- [15] Tensor product kernels for independence. DataSig Seminar, Mathematical Institute, University of Oxford, presentation (45 minutes), 26 May 2022.
- [16] When kernel machines meet shape constraints. Machine Learning External Seminar at the Gatsby Unit, UCL, presentation (50 minutes), 26 January 2022.
- [17] Continuous emotion transfer using RKHSs. Lifting Inference with Kernel Embeddings winter school and workshop (LIKE22; Bern, Switzerland), presentation (45 minutes), 12 January 2022.
- [18] Vector-valued infinite task learning in style transfer. CMStatistics, Advanced Statistical Methods for High Dimensional Data session, presentation (25 minutes), 19 December 2021.
- [19] Shape constraints meet kernel machines. Data Science Seminar at Eurecom, presentation (50 minutes), 4 November 2021.
- [20] Kernel regression with hard shape constraints. Workshop on Advances in Convex Optimization (EUROPT), presentation (25 minutes), 8 July 2021.
- [21] Vector-valued prediction with RKHSs and hard shape constraints. Computer Science and Systems Laboratory, Aix-Marseille University, presentation (1 hour), 20 May 2021.
- [22] Information theory, kernels and applications. Department of Statistics, LSE, presentation (50 minutes), 4 March 2021.

- [23] Shape-constrained kernel machines. Department of Statistics, Texas A&M University, presentation (1 hour), 12 February 2021.
- [24] Kernel machines with hard shape constraints. Meeting on Mathematical Statistics (MMS): Robustness and Computational Efficiency of Algorithms in Statistical Learning, presentation (40 minutes), 15 December 2020.
- [25] Kernel information theory and finance. D. E. Shaw Group, New York, U.S., presentation (30 minutes), 21 January 2020.
- [26] Towards large-scale approximation of tasks with derivatives – a kernel perspective. International Conference on Modern Mathematical Methods and High Performance Computing in Science & Technology (M3HPCST), Ghaziabad, India, presentation (30 minutes), 9-11 January 2020.
- [27] Orlicz Fourier features. International Indian Statistical Association Conference (IISA), IIT Bombay, India, presentation (20 minutes), 26-30 December 2019.
- [28] Consistency of Orlicz random Fourier features. EPFL, presentation (45 minutes), 23 September 2019.
- [29] Orlicz random Fourier features. Gatsby 21st Birthday Symposium, London, UK, presentation (20 minutes), 11-13 July 2019.
- [30] Outlier-robust divergence estimation on kernel-endowed domains with median of means. Third International Conference on Mathematics of Data Science (MathoDS 3), City University of Hong Kong (CityU), Hong Kong, China, presentation (30 minutes), 19-23 June 2019.
- [31] Towards outlier-robust statistical inference on kernel-enriched domains. RIKEN AIP Workshop, Tokyo, Japan, presentation (30 minutes), 15 April 2019.
- [32] From distance covariance to Hilbert-Schmidt independence criterion. Statistical Seminar in Rennes, presentation (1 hour), 26 October 2018.
- [33] HSIC, a measure of independence? Laboratory for Information and Inference Systems (LIONS), EPFL, presentation (40 minutes), 28 February 2018.
- [34] HSIC, an independence measure? Machine Learning & Computational Biology Lab, Department of Biosystems Science and Engineering (D-BSSE), ETH Zürich, presentation (1 hour), 26 February 2018.
- [35] Linear-time divergence measures with applications in hypothesis testing. Tao Seminar, INRIA Saclay, presentation (45 minutes), 13 February 2018.
- [36] Characterizing independence with tensor product kernels. Department of Statistics, Pennsylvania State University, presentation (1 hour), 13 December 2017.
- [37] Tensor product kernels: Independence and beyond. Google Brain, Mountain View, presentation (1 hour), 1 December 2017.
- [38] Tensor product kernels: Characteristic property and beyond. Advanced Methods Group, Cubist Systematic Strategies, New York, presentation (90 minutes), 28 November 2017.
- [39] Independence with tensor product kernels. Yahoo Research, New York, presentation (1 hour), 28 November 2017.
- [40] Tensor product kernels: Characteristic property and universality. Research Seminar, Sfs, ETH Zürich, presentation (45 minutes), 3 November 2017.
- [41] Characteristic tensor kernels. CREST Statistics Seminar, ENSAE, presentation (75 minutes), 9 October 2017.

- [42] Data-efficient independence testing with analytic kernel embeddings. PASADENA Seminar, Télécom ParisTech, presentation (1 hour), 17 May 2017.
- [43] Distribution regression: A simple technique with minimax-optimal guarantee. Parisian Statistics Seminar, Henri Poincaré Institute, presentation (1 hour), 27 March 2017.
- [44] A linear-time adaptive nonparametric two-sample test. Signal Processing and Machine Learning Seminar, Marseilles, presentation (1 hour), 24 March 2017.
- [45] Minimax-optimal distribution regression. Probability and Statistics Seminar, Orsay, presentation (1 hour), 16 March 2017.
- [46] T-testing: A linear-time adaptive nonparametric technique. Machine Learning Seminar, Télécom ParisTech, presentation (1 hour), 2 February 2017.
- [47] Distribution regression. New Directions for Learning with Kernels and Gaussian Processes Dagstuhl Seminar, presentation (30 minutes), 1 December 2016.
- [48] Adaptive linear-time nonparametric t-test. Facebook AI Research, Paris, France, presentation (45 minutes), 21 November 2016.
- [49] Distinguishing distributions with maximum testing power. Realeyes, Budapest, Hungary, presentation (1 hour), 24 August 2016.
- [50] Hypothesis testing with kernels. International Workshop on Pattern Recognition in Neuroimaging (PRNI), Trento, Italy, presentation (1 hour), 22-24 June 2016.
- [51] Kernel-based learning on probability distributions. University of California, San Diego, presentation (30 minutes), 25 April 2016.
- [52] Distribution regression with minimax-optimal guarantee. MASCOT-NUM, presentation (45 minutes), 25 March 2016.
- [53] Performance guarantees for kernel-based learning on probability distributions. Special Symposium on Intelligent Systems, MPI, Tübingen, presentation (20 minutes), 16 March 2016.
- [54] Optimal rates for the random Fourier feature technique. École Polytechnique, presentation (2 hours), 14 March 2016.
- [55] Learning theory for vector-valued distribution regression. CMStatistics 2015, presentation (35 minutes), 12 December 2015.
- [56] Optimal uniform and  $L^p$  rates for random Fourier features. Pennsylvania State University, presentation (1 hour), 4 December 2015.
- [57] Optimal rates for the random Fourier feature method. Statistical ML Reading Group, Carnegie Mellon University, presentation (1 hour), 1 December 2015.
- [58] Distribution regression: Computational and statistical tradeoffs. ML Lunch Seminar, Carnegie Mellon University, presentation (50 minutes), 30 November 2015.
- [59] Distribution regression: Computational and statistical tradeoffs. Princeton University, presentation (1 hour), 26 November 2015.
- [60] Optimal rates for random Fourier feature approximations. University of Alberta, presentation (1 hour), 24 November 2015.
- [61] Optimal rates for random Fourier feature kernel approximations. AMPLab, UC Berkeley, presentation (1 hour), 20 November 2015.

- [62] Performance guarantees for random Fourier features - limitations and merits. Neil Lawrence's lab, University of Sheffield, presentation (1 hour), 25 June 2015.
- [63] Regression on probability measures: A simple and consistent algorithm. Centre for Research in Statistical Methodology Seminars, Department of Statistics, University of Warwick, presentation (1 hour), 29 May 2015.
- [64] Vector-valued distribution regression - keep it simple and consistent. Computational Statistics and Machine Learning reading group, Department of Statistics, University of Oxford, presentation (50 minutes), 1 May 2015.
- [65] A simple and consistent technique for vector-valued distribution regression. Artificial Intelligence and Natural Computation seminars, University of Birmingham, presentation (50 minutes), 26 January 2015.
- [66] Consistent vector-valued regression on probability measures. Bernhard Schölkopf's Lab, MPI for Intelligent Systems, Tübingen, presentation (1 hour), 15 January 2015.
- [67] Consistent distribution regression via mean embedding. University of Hertfordshire, presentation (1 hour), 5 March 2014.
- [68] Dictionary learning: Independence, structured sparsity and beyond. Gatsby Unit, UCL, presentation (45 minutes), 23 April 2013.
- [69] Beyond independent subspace analysis. INRIA, SIERRA project-team, presentation (90 minutes), 17 January 2012.
- [70] Hedging with Lasso. Morgan Stanley, presentation (25 minutes), 9 September 2011.
- [71] Structured sparsity and non-convex sparsity-inducing methods. Morgan Stanley, presentation (25 minutes), 9 May 2011.
- [72] Online group-structured dictionary learning. Machine Learning at Budapest, presentation (45 minutes), 22 November 2010.
- [73] Analysis and prediction of time series with missing data. Morgan Stanley, Speaker Series Event, presentation (30 minutes), 9 October 2009.
- [74] Analysis and prediction of time series with missing data. Morgan Stanley - BME Financial Innovation Centre Kick-off & Workshop, presentation (25 minutes), 15 June 2009.
- [75] Exploration of behavioral patterns and its applications in human-computer interaction. Info Savaria, Szombathely, presentation (30 minutes), 14-16 April 2005.
- [76] Recognition of behavioral patterns and its potentials of human-computer interaction. Info ÉRA, Békéscsaba, presentation (30 minutes), 14-16 April 2004.

#### INVITED TALKS & POSTERS (LOCAL)

- [1] The minimax rate of HSIC estimation. LSE Statistics Research Showcase, Department of Statistics, LSE, presentation (25 minutes), 20 June 2024.
- [2] Kernelized cumulants: Beyond mean embeddings. LSE Statistics Research Showcase, Department of Statistics, LSE, presentation (25 minutes), 5 June 2023.
- [3] Kernel machines with shape constraints. PhD Open Day, Department of Statistics, LSE, poster, 28 November 2022.

- [4] Support vector machines with hard shape constraints. Combinatorics, Game Theory, and Optimisation (CGO) Seminar, Department of Mathematics, LSE, presentation (50 minutes), 29 September 2022.
- [5] Shape-constrained kernel machines. Data Science Research Lightning Talks event, Data Science Institute, LSE, presentation (3 minutes), 21 September 2022.
- [6] Continuous emotion transfer. LSE Statistics Research Showcase, Department of Statistics, LSE, presentation (25 minutes), 15 June 2022.
- [7] Information theory, kernels & applications. Research Showcase at Data Science Institute, LSE, presentation (5 minutes), 13 December 2021.
- [8] Distribution regression and beyond. LSE Statistics Open House, Department of Statistics, LSE, presentation (20 minutes), 14 October 2021.
- [9] Applications of kernel-based information theoretical measures. LSE Statistics Open House, Department of Statistics, LSE, presentation (15 minutes), 14 October 2021.
- [10] Optimal regression on sets. UCL eResearch Domain launch event, UCL, London, UK, poster, 29 June 2016.
- [11] Vector-valued distribution regression: A simple and consistent approach. Statistical Science Seminars, UCL, presentation (1 hour), 9 October 2014.
- [12] Distribution regression - the set kernel heuristic is consistent. CSML Lunch Talk Series, UCL, presentation (1 hour), 2 May 2014.
- [13] Dictionary optimization problems and their applications. Eötvös Loránd University, Day of Science, presentation (40 minutes), 22 November 2012.
- [14] Recommender systems, applications in education. Child's Play with Adult's Mind, Conference, Budapest University of Technology and Economics, presentation (20 minutes), 22 March 2012.
- [15] Collaborative filtering via group-structured dictionary learning. Eötvös Loránd University, Innovation Day, poster, 23 February 2012.
- [16] Interpreting natural language: applications of group-structured dictionary learning. Eötvös Loránd University, Faculty of Informatics, Neumann's Day, poster, 12 May 2011.
- [17] Interpreting words: an application of (structured) sparse coding. Eötvös Loránd University, Faculty of Informatics, Neumann's Day, poster, 12 May 2011.
- [18] Online group-structured dictionary learning. Eötvös Loránd University, Faculty of Informatics, Neumann's Day, poster, 12 May 2011.
- [19] Online group-structured dictionary learning. Eötvös Loránd University, TÁMOP Research Seminar, presentation (45 minutes), 28 January 2011.
- [20] Online structured dictionary learning and its applications. Eötvös Loránd University, Problem Solving Seminar for Applied Mathematicians, presentation (45 minutes), 5 November 2010.
- [21] Nonparametric regression, Lasso. Eötvös Loránd University, Problem Solving Seminar for Applied Mathematicians, presentation (45 minutes), 12 November 2009.
- [22] Independent subspace analysis; tensor-SVD, tensorfaces; blind subspace deconvolution. Eötvös Loránd University, Problem Solving Seminar for Applied Mathematicians, presentation (45 minutes), 19 October 2007.

- [23] Adaptive human-computer interaction via face and gaze tracking. Eötvös Loránd University, Faculty of Informatics, Neumann’s Day, presentation (20 minutes), 6 November 2003.

## SOFTWARE

<https://zoltansz.github.io/software.html>

## LECTURING (**P** – PHD, **G** – GRADUATE, **U** – UNDERGRADUTE, **F** – FOR FUN)

**LSE** (UK; WT = Winter Term, AT = Autumn Term)

- Foundations of Machine Learning (ST510): Kernel Methods (**p**) WT (TBA), 2026/27
  - lecture (2h; TBA) and practical session (1.5h; TBA).
- Graph Data Analytics and Representation Learning (ST457; **g**) WT, 2026/27
  - lecturing and practical sessions.
- Machine Learning and Data Mining (ST443; **g**) AT, 2026/27
  - practical sessions: Phil Chan (3 seminar groups), Kaifang Zhou (1 seminar group).
- Getting Started with Linux (**f**) Sept. 25, 2026
  - mini-course (2.5 hours).
- Foundations of Machine Learning (ST510): Kernel Methods (**p**) WT (Feb. 9, Feb. 12), 2025/26
  - lecture (2h; Feb. 9) and practice session (1.5h; Feb. 12).
- Graph Data Analytics and Representation Learning (ST457; **g**) WT, 2025/26
  - 28 students,
  - lecturing and practical sessions.
- Machine Learning (ST310; **u**) AT, 2025/26
  - practical sessions: Phil Chan (3 seminar groups),  
Seyedpouya Mirrezaeiroudaki (3), Erhan Xu (1),
  - 141 students.
- Getting Started with Linux (**f**) Sept. 26, 2025
  - mini-course (2.5 hours),
  - 42 registered participants (BSc, MSc, PhD).
- Foundations of Machine Learning (ST510): Kernel Methods (**p**) WT (Feb. 10), 2024/25
- Graph Data Analytics and Representation Learning (ST457; **g**) WT, 2024/25
  - 22 students,
  - practical sessions: Shakeel Gavioli-Akilagun.
- Linux – The Operating System of Freedom (**f**) Sept. 27, 2024
  - mini-course (2.5 hours),
  - 58 registered participants (BSc, MSc, PhD, Staff).
- Foundations of Machine Learning (ST510): Support Vector Machines (**p**) WT (Feb. 5), 2023/24
- Graph Data Analytics and Representation Learning (ST457; **g**) WT, 2023/24
  - 17 students,
  - practical sessions: Shakeel Gavioli-Akilagun.
- Linux – The Operating System of Freedom (**f**) Sept. 22, 2023
  - mini-course (2.5 hours),
  - 36 registered participants (MSc).
- Linux – The Operating System of Freedom (**f**) May 23, 2023
  - mini-course (2.5 hours),
  - 15 registered participants (PhD, PostDoc, Staff).
- Graph Data Analytics and Representation Learning (ST457; **g**) WT, 2022/23
  - 30 students,
  - lecturing and practical sessions.
- Artificial Intelligence (**g**) AT, 2022/23
  - ca. 50 students,
  - practical sessions: Francesca Panero, Marcos Barreto.

|                                                                                         |                    |
|-----------------------------------------------------------------------------------------|--------------------|
| Artificial Intelligence ( <b>g</b> )                                                    | AT, 2021/22        |
| – ca. 25 students,                                                                      |                    |
| – practical sessions: Marcos Barreto.                                                   |                    |
| <b>École Polytechnique</b> (France)                                                     |                    |
| Independence Measures and Testing ( <b>g</b> )                                          | Spring, 2021       |
| – mini-course (4 hours).                                                                |                    |
| Structured Data: Learning, Prediction, Dependency, Testing ( <b>g</b> , M2 MDA)         | Spring, 2019       |
| – ca. 60 students,                                                                      |                    |
| – with Prof. Florence d’Alché-Buc, Prof. Slim ESSID.                                    |                    |
| Advanced Machine Learning ( <b>g</b> , X-HEC)                                           | Spring, 2019       |
| – ca. 60 students,                                                                      |                    |
| – with Prof. Stéphane Canu, Prof. Erwan Le Pennec, Thomas Kerdeux,                      |                    |
| – tutoring: Jaouad Mourtada, Nicolas Prost.                                             |                    |
| Statistics ( <b>g</b> , X-HEC)                                                          | Fall, 2018         |
| – ca. 60 students,                                                                      |                    |
| – with Prof. Stéphanie Allasonnière, Prof. Elodie Vernet, Geneviève Robin, Wei Jiang.   |                    |
| Introduction to Machine Learning ( <b>g</b> , X-HEC)                                    | Fall, 2018         |
| – ca. 60 students,                                                                      |                    |
| – with Prof. Julie Josse, Prof. Erwan Scornet, Prof. Sylvain Le Corff, Florian Bourgey. |                    |
| Structured Data: Learning, Prediction, Dependency, Testing ( <b>g</b> , M2 MDA)         | Spring, 2018       |
| – ca. 70 students,                                                                      |                    |
| – with Prof. Florence d’Alché-Buc, Prof. Slim ESSID, Prof. Arthur Tenenhaus,            |                    |
| Alexandre Garcia (Datalab).                                                             |                    |
| Advanced Machine Learning ( <b>g</b> , X-HEC)                                           | Spring, 2018       |
| – ca. 60 students,                                                                      |                    |
| – with Prof. Stéphane Canu, Prof. Erwan Le Pennec, Anne Auger.                          |                    |
| Structured Data: Learning, Prediction, Dependency, Testing ( <b>g</b> , M2 MDA)         | Spring, 2017       |
| – ca. 95 students,                                                                      |                    |
| – with Prof. Florence d’Alché-Buc, Prof. Slim ESSID, Prof. Arthur Tenenhaus.            |                    |
| Functional Data Analysis ( <b>g</b> )                                                   | Fall, 2016         |
| – special course.                                                                       |                    |
| <b>INRIA</b> (France)                                                                   |                    |
| Kernel Methods, Divergence and Independence Measures, Hypothesis Testing ( <b>g</b> )   | July 15–17, 2019   |
| – @ Summer School on Data Science for Document Analysis                                 | July 18 & 20, 2018 |
| and Understanding,                                                                      |                    |
| – 6-hour long course / year.                                                            |                    |
| Manifold Learning and Classification for EEG Analysis ( <b>g</b> )                      | July 27, 2017      |
| – @ Summer School on Mathematical and Computational Methods,                            |                    |
| for Life Sciences,                                                                      |                    |
| – 3-hour long course.                                                                   |                    |
| <b>HEC Paris</b> (France)                                                               |                    |
| Data for Management Certificate ( <b>g</b> )                                            | May 7 & 10, 2019   |
| – with Prof. Karim Lounici, Prof. Sylvain Le Corff,                                     |                    |
| – 8 hours / day.                                                                        |                    |
| <b>Carnegie Mellon University</b> (US)                                                  |                    |
| Kernel-based Dependency Measures and Hypothesis Testing ( <b>g</b> )                    | Nov. 27, 2017      |
| – Guest Lecture @ School of Computer Science,                                           |                    |
| – 80 minutes.                                                                           |                    |
| <b>University College London</b> (UK)                                                   |                    |
| Advanced Topics in Machine Learning - Theory of RKHS ( <b>g</b> )                       | Spring, 2015–2016  |

- ca. 60 students,
  - with Prof. Arthur Gretton, Kacper Chwialkowski.
- Adaptive Modelling, Introduction to Kernel Methods (**g**) Spring, 2015–2016
- ca. 20 students,
  - with Prof. Arthur Gretton, Heiko Strathmann, Wittawat Jitkrittum.

#### **Eötvös Loránd University** (Hungary)

- Reinforcement Learning (**g**) Spring, 2009–2013
- ca. 45 students in each semester,
  - with Prof. András Lőrincz.
- Artificial Neural Networks (**g**) Fall, 2008–2012
- ca. 45 students in each semester,
  - with Prof. András Lőrincz.
- Image Processing, Speech Recognition, Applications of Artificial Intelligence (**g**) 2007–2008
- ca. 25 students in each semester.
- Introduction to Mathematics (**u**) 2006–2007
- ca. 25 students in each semester.
- Symbolic Programming (**u**) 2004–2006
- ca. 25 students in each semester.

#### SUPERVISION

- Thomas Borsani (Ph.D.) Aug. 3, 2026–
- Free University of Bozen-Bolzano, Italy,
  - Ph.D. visitor.
- Hanqi Wang (Ph.D.) Sept., 2025–
- Department of Statistics, LSE, UK,
  - co-supervised with Prof. Tengyao Wang,
  - role: secondary supervisor.
- Jose Cribeiro-Ramallo (Ph.D.) Aug., 2025–
- Faculty of Computer Science, Karlsruhe Institute of Technology, Germany,
  - co-supervised with Prof. Thorsten Strufe,
  - role: secondary supervisor.
- Tao Ma (Ph.D.) Jan., 2024–Sept., 2025
- Department of Statistics, LSE, UK,
  - co-supervised with Prof. Milan Vojnovic,
  - role: primary supervisor,
  - topic: Trustworthy Decision Making with Sustainability.
- Xiaoyi Wen (Ph.D.) Sept., 2023–Aug., 2024
- Renmin University of China, China,
  - Ph.D. visitor awarded by CSC (China Scholarship Council),
  - topic: Joint Independence Testing for Latent Position Random Graphs.
- Florian Kalinke (Ph.D.) Sept. – Dec., 2022
- Faculty of Computer Science, Karlsruhe Institute of Technology, Germany,
  - Ph.D. visitor,
  - topic: Accelerated Kernel-Based Information Theoretical Estimators.
- Pingfan Su (Ph.D.) Sept., 2022–
- Department of Statistics, LSE, UK,
  - co-supervised with Prof. Chengchun Shi,
  - role: secondary supervisor,
  - topic: Reinforcement Learning in Nonstationary Environment.
- Sakina Hansen (Ph.D.) Sept., 2022 – Aug., 2023
- Department of Statistics, LSE, UK,

|                                                                                                                                                                                                                                                                                                                                  |                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <ul style="list-style-type: none"> <li>– co-supervised with Prof. Joshua Loftus,</li> <li>– role: secondary supervisor,</li> <li>– topic: Causality, Counterfactuals and Meta-learning to Address the Complexity of Fairness in Data Science and Machine Learning.</li> </ul>                                                    |                         |
| Linda Chamakh (Ph.D.)                                                                                                                                                                                                                                                                                                            | Apr., 2018 – May, 2021  |
| <ul style="list-style-type: none"> <li>– CMAP, École Polytechnique &amp; BNP Paribas, France,</li> <li>– co-supervised with Prof. Emmanuel Gobet, Jean-Philippe Lemor,</li> <li>– topic: Uncertainty Quantification, Robustness of Systematic Strategies.</li> </ul>                                                             |                         |
| Alex Lambert (Ph.D.)                                                                                                                                                                                                                                                                                                             | Oct., 2017 – May, 2021  |
| <ul style="list-style-type: none"> <li>– CMAP, École Polytechnique &amp; LTCI, Télécom ParisTech, France,</li> <li>– co-supervised with Prof. Florence d’Alché-Buc,</li> <li>– topic: Statistical Learning of Vector-Valued Functions with Operator Random Fourier Features.</li> </ul>                                          |                         |
| Gaspar Massiot (PostDoc)                                                                                                                                                                                                                                                                                                         | Oct., 2017 – Aug., 2018 |
| <ul style="list-style-type: none"> <li>– French Aerospace Lab ONERA, France,</li> <li>– co-supervised with Prof. Éric Moulines, Sidonie Lefebvre,</li> <li>– topic: Kernel Methods in Hyperspectral Imaging.</li> </ul>                                                                                                          |                         |
| Romain Brault (PostDoc)                                                                                                                                                                                                                                                                                                          | Oct., 2017 – Oct., 2018 |
| <ul style="list-style-type: none"> <li>– CMAP, École Polytechnique &amp; LTCI, Télécom ParisTech, France,</li> <li>– co-supervised with Prof. Florence d’Alché-Buc, Prof. Arthur Tenenhaus,</li> <li>– topic: Prediction of Functional Outputs by Kernels.</li> </ul>                                                            |                         |
| Zoltán Milacski (M.Sc.)                                                                                                                                                                                                                                                                                                          | 2012–2013               |
| <ul style="list-style-type: none"> <li>– School of Computer Science, Eötvös Loránd University (ELU), Hungary,</li> <li>– co-supervised with Prof. András Lőrincz of the national student competitor (2<sup>nd</sup> prize),</li> <li>– topic: Recurrent Reinforcement Learning in High-Frequency Algorithmic Trading.</li> </ul> |                         |
| Gabriella Merész (M.Sc.)                                                                                                                                                                                                                                                                                                         | 2012                    |
| <ul style="list-style-type: none"> <li>– Department of Applied Mathematics, ELU, Hungary,</li> <li>– topic: Prediction of Financial Time Series via ARMA-GARCH Methods.</li> </ul>                                                                                                                                               |                         |

## MENTORING

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Capstone project (M.Sc.)                                                                                                                                                                                                                                                                                                                       | Nov., 2025 – Aug, 2026 |
| <ul style="list-style-type: none"> <li>– Department of Statistics, LSE, UK,</li> <li>– team members: Anastasiia Prokhorova, Keshni Shah, Sofia Jain,</li> <li>– topic: Early Indicators of Gentrification,</li> <li>– joint mentoring with Anton Boychenko (Transport for London).</li> </ul>                                                  |                        |
| Capstone project (M.Sc.)                                                                                                                                                                                                                                                                                                                       | Nov., 2025 – Aug, 2026 |
| <ul style="list-style-type: none"> <li>– Department of Statistics, LSE, UK,</li> <li>– team members: Jaey Tay, Norah Kuduk, Ritika Annapareddy,</li> <li>– topic: Delivering Elite European Football (Soccer) Analytics,</li> <li>– joint mentoring with Mohammad Ashkani and Mateusz Faltyn (Trilemma Foundation).</li> </ul>                 |                        |
| Alessandro De Palma (Assistant Professor)                                                                                                                                                                                                                                                                                                      | Nov., 2025 –           |
| <ul style="list-style-type: none"> <li>– Department of Statistics, LSE, UK,</li> <li>– academic mentoring of our junior colleague.</li> </ul>                                                                                                                                                                                                  |                        |
| Capstone project (M.Sc.)                                                                                                                                                                                                                                                                                                                       | Nov., 2024 – Aug, 2025 |
| <ul style="list-style-type: none"> <li>– Department of Statistics, LSE, UK,</li> <li>– team members: Lotte-Sophie Folbert, Chi Yunng Lim, Guillem Tobias-Larrauri,</li> <li>– topic: Quantifying the Impact of TfL’s Superloop Bus Network on House Prices,</li> <li>– joint mentoring with Anton Boychenko (Transport for London).</li> </ul> |                        |
| Capstone project (M.Sc.)                                                                                                                                                                                                                                                                                                                       | Nov., 2024 – Aug, 2025 |
| <ul style="list-style-type: none"> <li>– Department of Statistics, LSE, UK,</li> </ul>                                                                                                                                                                                                                                                         |                        |

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| <ul style="list-style-type: none"> <li>– team members: Shiyu Wu, Yikun Jiang, Zoe He,</li> <li>– topic: Hierarchical optimization methods comparison for Causal AutoML,</li> <li>– joint mentoring with Egor Kraev (Wise).</li> </ul>                                                                                                                            | Nov., 2023 – July, 2024 |
| <p>Capstone project (M.Sc.)</p> <ul style="list-style-type: none"> <li>– Department of Statistics, LSE, UK,</li> <li>– team members: Maciej Kuczek, Reuben Mathew, Tweshaa Dewan,</li> <li>– topic: Statistical Methods in AB Testing,</li> <li>– joint mentoring with Egor Kraev and Julian Teichgraber (Wise).</li> </ul>                                      | Sept., 2023 – Aug. 2024 |
| <p>Christine Yuen, Francesca Panero, Marcos Barreto (Assistant Professors)</p> <ul style="list-style-type: none"> <li>– Department of Statistics, LSE, UK,</li> <li>– academic mentoring of our junior colleagues,</li> <li>– joint mentoring with Prof. Milan Vojnovic.</li> </ul>                                                                              | Nov., 2022 – Aug., 2023 |
| <p>Capstone project (M.Sc.)</p> <ul style="list-style-type: none"> <li>– Department of Statistics, LSE, UK,</li> <li>– team members: Jingyan Lu, Sizhe Li, Yujie Zhang, Zhendong Wang,</li> <li>– topic: Food Recognition and Nutritional Analysis from Food Images,</li> <li>– joint mentoring with Wittawat Jitkrittum (Google Research, New York).</li> </ul> | 2022–2023               |
| <p>Mentoring (B.Sc.)</p> <ul style="list-style-type: none"> <li>– 20 B.Sc. students in Actuarial Science, Mathematics, Statistics and Business,</li> <li>– academic mentoring.</li> </ul>                                                                                                                                                                        | 2021–2022               |
| <p>Mentoring (B.Sc.)</p> <ul style="list-style-type: none"> <li>– 12 B.Sc. students in Actuarial Science, Mathematics, Statistics and Business,</li> <li>– academic mentoring.</li> </ul>                                                                                                                                                                        | Dec., 2021 – Aug., 2022 |
| <p>Capstone project (M.Sc.)</p> <ul style="list-style-type: none"> <li>– Department of Statistics, LSE, UK,</li> <li>– team members: Deelan Gopaul, Josiah Suartono,</li> <li>– topic: Forecasting sea water transparency,</li> <li>– joint mentoring with Dima Karamshuk (Facebook).</li> </ul>                                                                 | Oct. – Dec., 2021       |
| <p>Jonathan Cardoso-Silva (Research Officer)</p> <ul style="list-style-type: none"> <li>– Department of Methodology, LSE, UK.</li> </ul>                                                                                                                                                                                                                         | May – Aug., 2020        |
| <p>Bechir Trabelsi (M.Sc.)</p> <ul style="list-style-type: none"> <li>– intern from ESTA, France,</li> <li>– topic: Shape-Constrained Risk Measures.</li> </ul>                                                                                                                                                                                                  | Oct., 2019 – Jan., 2020 |
| <p>Michaël Allouche (M.Sc.)</p> <ul style="list-style-type: none"> <li>– CMAP, École Polytechnique, France,</li> <li>– co-supervised with Prof. Emmanuel Gobet,</li> <li>– topic: Structured Dictionary Learning of Migration Matrices.</li> </ul>                                                                                                               | GANs on Time Series     |
| <p>Meyer Scetbon (M.Sc.)</p> <ul style="list-style-type: none"> <li>– intern from INRIA, France,</li> <li>– co-supervised with Gaël Varoquaux,</li> <li>– topic: Fast Kernel-based Hypothesis Testing.</li> </ul>                                                                                                                                                | Jan. – Mar., 2018       |
| <p>Flore Martin, Roulier Lorraine (M.Sc.)</p> <ul style="list-style-type: none"> <li>– Sciences for Environmental Challenges, École Polytechnique, France,</li> <li>– topic: Low-Dimensional Embedding of Environmental Variables.</li> </ul>                                                                                                                    | Fall, 2017              |
| <p>Moussab Djerrab (Ph.D.)</p> <ul style="list-style-type: none"> <li>– LTCI, Télécom ParisTech, France,</li> <li>– topic: Structured Prediction with Surrogate Losses.</li> </ul>                                                                                                                                                                               | 2013–2016               |
| <p>Wittawat Jitkrittum (Ph.D.)</p> <ul style="list-style-type: none"> <li>– Gatsby Unit, University College London, UK,</li> <li>– topic: Kernel Techniques, Statistics.</li> </ul>                                                                                                                                                                              | 2012–2013               |
| <p>Máté Csákvári, Zoltán Tóssér (M.Sc.)</p>                                                                                                                                                                                                                                                                                                                      |                         |

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| <ul style="list-style-type: none"> <li>– School of Computer Science, ELU, Hungary,</li> <li>– topic: Information Theory, Dictionary Learning.</li> </ul>                                         | 2011–2013                 |
| László Jeni (PostDoc)                                                                                                                                                                            |                           |
| <ul style="list-style-type: none"> <li>– Robotics Institute, Carnegie Mellon University, US,</li> <li>– topic: Extensions of Constrained Local Models, Facial Expression Recognition.</li> </ul> | 2011–2013                 |
| Balázs Pintér, Gyula Vörös (Ph.D.)                                                                                                                                                               |                           |
| <ul style="list-style-type: none"> <li>– School of Computer Science, ELU, Hungary,</li> <li>– topic: Structured-Sparse Coding and Dictionary Learning in Natural Language Processing.</li> </ul> | 2011–2013                 |
| András Sárkány (M.Sc.)                                                                                                                                                                           |                           |
| <ul style="list-style-type: none"> <li>– School of Computer Science, ELU, Hungary,</li> <li>– topic: Hedging via Sparse Coding.</li> </ul>                                                       | 2011 Autumn – 2012 Spring |
| Gergő Hammer (M.Sc.)                                                                                                                                                                             |                           |
| <ul style="list-style-type: none"> <li>– Department of Applied Mathematics, ELU, Hungary,</li> <li>– topic: Self-Similar Structures for Financial Prediction.</li> </ul>                         | 2009 Autumn – 2010 Spring |
| Dávid Retek, Mária Mészáros (M.Sc.)                                                                                                                                                              |                           |
| <ul style="list-style-type: none"> <li>– Department of Applied Mathematics, ELU, Hungary,</li> <li>– topic: Online Structured Dictionary Learning and Its Applications.</li> </ul>               | 2007 Autumn – 2008 Spring |
| Gabriella Merész, Kitti Korbács, Nóra Villányi (M.Sc.)                                                                                                                                           |                           |
| <ul style="list-style-type: none"> <li>– Department of Applied Mathematics, ELU, Hungary,</li> <li>– topic: Tensor Textures.</li> </ul>                                                          | 2007 Autumn – 2008 Spring |
| Anikó Márton, Kata Péter (M.Sc.)                                                                                                                                                                 |                           |
| <ul style="list-style-type: none"> <li>– Department of Applied Mathematics, ELU, Hungary,</li> <li>– topic: Temporal Independent Subspace Analysis of Facial Features.</li> </ul>                |                           |

## Referent Professor

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| Nicolas Bonnet's internship (M.Sc.)                                                                                                                                                                                              | Mar. – July, 2019  |
| <ul style="list-style-type: none"> <li>– École Polytechnique &amp; HEC Paris, France,</li> <li>– internship @ Atos (India),</li> <li>– title: Vertical Integration of Supply and Delivery Chain from India to France.</li> </ul> |                    |
| Camille Jandot's internship (M.Sc.)                                                                                                                                                                                              | Apr. – Sept., 2017 |
| <ul style="list-style-type: none"> <li>– Télécom ParisTech, France,</li> <li>– title: Modelling Space Time Series with Operator-Valued Kernels</li> <li>– Application to Detection of Epidemics.</li> </ul>                      |                    |

## GRANTS

|                                                                                                                                                                                                                                                                                                                                                                     |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| LSE Global Research Fund                                                                                                                                                                                                                                                                                                                                            | 2023-2024 |
| <ul style="list-style-type: none"> <li>topic: Computational vs. Statistical Analysis of Kernel Stein Discrepancy,</li> <li>joint work with: <ul style="list-style-type: none"> <li>Prof. Bharath K. Sriperumbudur (Pennsylvania State University),</li> <li>Florian Kalinke (Karlsruhe Institute of Technology),</li> </ul> </li> <li>amount: 5,000 GBP.</li> </ul> |           |
| Europlace Institute of Finance (EIF)                                                                                                                                                                                                                                                                                                                                | 2020–2021 |
| <ul style="list-style-type: none"> <li>topic: Machine Learning for Risk Management: Kernels with Shape Constraints,</li> <li>joint work with: <ul style="list-style-type: none"> <li>Prof. Dino Sejdinovic (University of Oxford),</li> <li>Olivier Derollez (BNP Paribas),</li> </ul> </li> <li>amount: 10,000 EUR.</li> </ul>                                     |           |
| Labex DigiCosme                                                                                                                                                                                                                                                                                                                                                     | 2017–2018 |
| <ul style="list-style-type: none"> <li>keywords: operator-valued kernels, prediction of function-valued functions,</li> <li>joint work with: <ul style="list-style-type: none"> <li>Prof. Florence d'Alché-Buc (Télécom Paris),</li> </ul> </li> </ul>                                                                                                              |           |

Prof. Arthur Tenenhaus (CentraleSupélec),  
amount: 50,000 EUR.

## ACADEMIC AWARDS AND RESEARCH SCHOLARSHIPS

|                                                                                |           |
|--------------------------------------------------------------------------------|-----------|
| Winton Prize for the best Capstone Project                                     | 2023      |
| – topic: Forecasting sea water transparency                                    |           |
| – team: Deelan Gopaul, Josiah Suartono,                                        |           |
| – joint supervision with Dima Karamshuk (Facebook).                            |           |
| Best Paper Award at NeurIPS (awarded to 3 papers out of the 3240 submissions)  | 2017      |
| Research Scholarship of the John von Neumann Computer Society                  | 2005–2012 |
| Bronze Medal of the Pro Patria et Scientia Award of Hungarian Ph.D. Students   | 2008      |
| Scientist of the Year Award of the School of Computer Science                  | 2007      |
| Research Scholarship of the Bliss Foundation                                   | 2004      |
| Outstanding Student Award of the Faculty of Natural Sciences                   | 2003      |
| Research Scholarship of the Eötvös Loránd University                           | 2003      |
| National Scientific Student Competition and Conference (2 <sup>nd</sup> prize) | 2002      |

## INTERVIEW

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|-----------------------------------------------------------------|---------------|
| HCI study on AI assistance for ICLR ACs                         | Aug. 30, 2023 |
| Memory (film; topic: AI & Brain)                                | Dec., 2018    |
| At TWiML & AI on our work winning best paper award at NIPS-2017 | Dec., 2017    |

## LANGUAGES

- English (fluent), Spanish (basic), Hungarian (native).
- Computer languages: Python, Matlab/Octave, Maple, L<sup>A</sup>T<sub>E</sub>X, HTML.